



The Ekert protocol

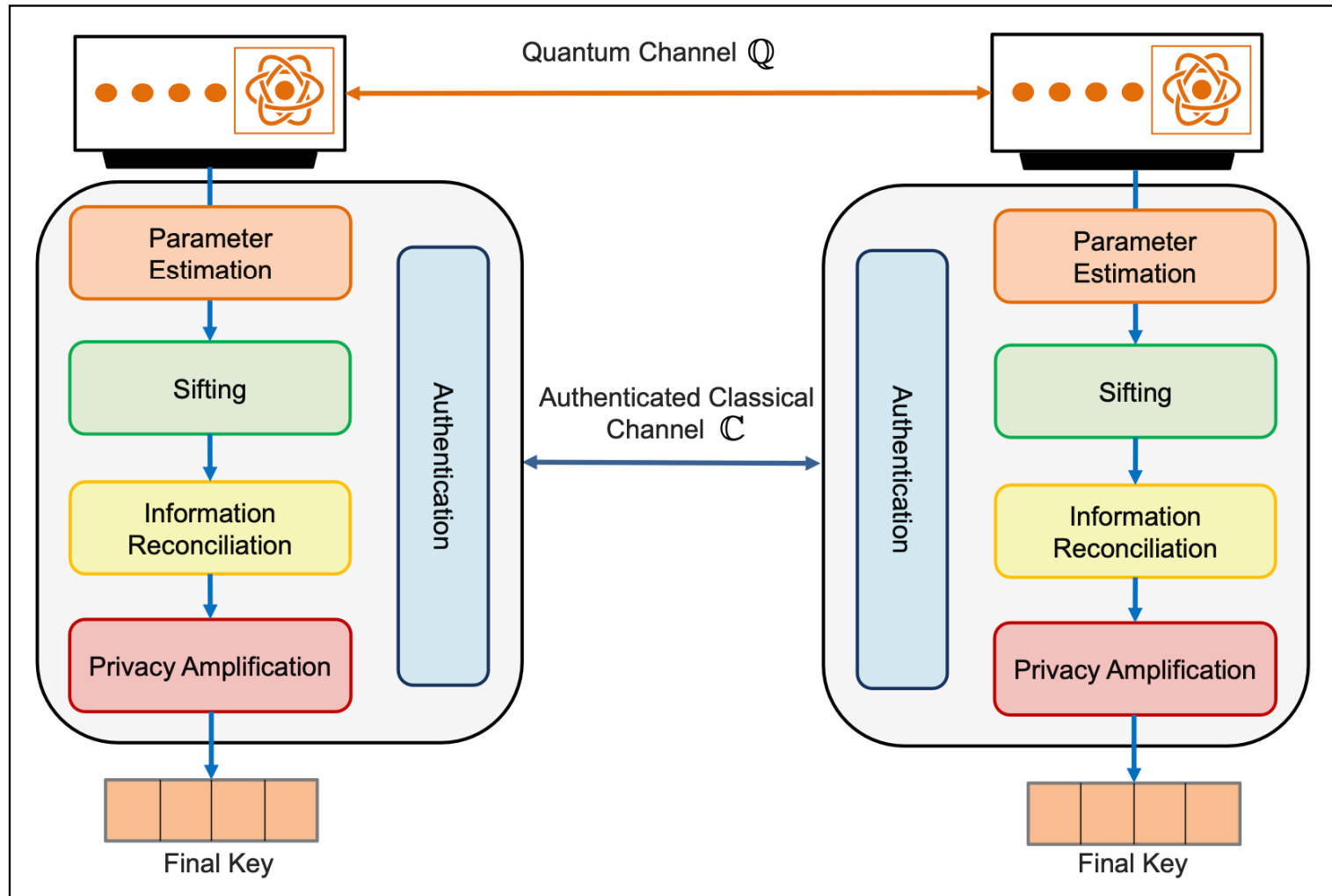
1. Alice and Bob distribute a number of $|\Psi^-\rangle_{AB}$ states between them.
2. For each state, Alice and Bob randomly choose a measurement from the sets $\{A_i\}$ and $\{B_i\}$.
3. Alice and Bob announce the bases they chose for each measurement. The results of the pairs (A_1, B_1) and (A_3, B_3) form the sifted key.
4. The results for the pairs (A_1, B_3) , (A_1, B_2) , (A_2, B_3) and (A_2, B_2) are used to check the CHSH inequality.
5. If CHSH test passed:
 - (i) Proceed with **error correction**.
 - (ii) Apply **privacy amplification** to eliminate Eve's potential information.
 - (iii) Result: a **shared secret key**.
6. Entanglement guarantees security even if source is untrusted.

Overview



QKD Post Processing

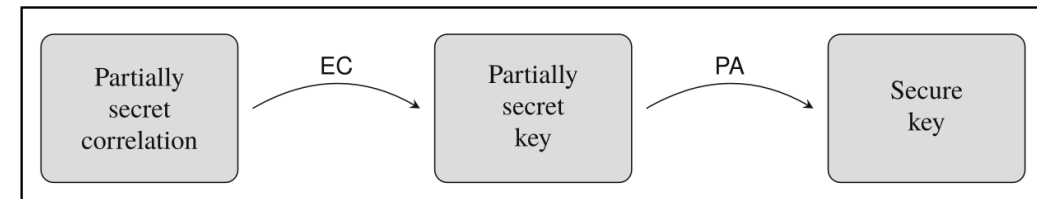
QKD Classical Post-Processing





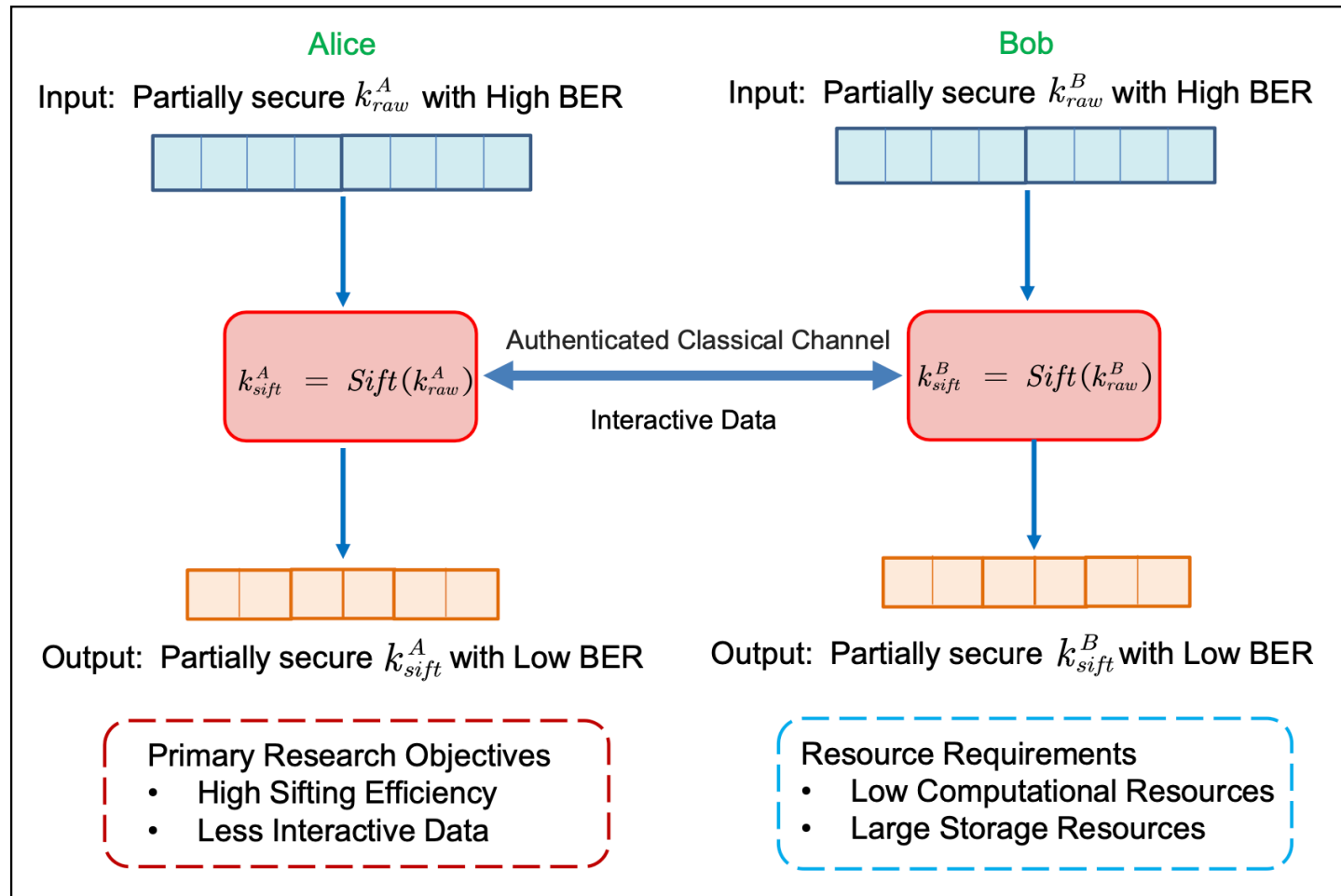
Classical Post-Processing

- After the quantum transmission phase, Alice and Bob hold bit strings that are, in general, partially correlated and partially secure.
- **Goal:** Turn these into secure, identical keys via classical post-processing.
- **Step 1: Parameter Estimation:**
 - Estimate the error rate between Alice and Bob's bit strings.
 - Infer how much information might have leaked to Eve.
 - Abort the protocol if the error rate is too high.
- **Step 2: Error Correction (EC):**
 - Correct errors to ensure Alice and Bob share identical bit strings.
- **Step 3: Privacy Amplification (PA)**
 - Reduce Eve's residual information to negligible.
 - Output: A short, but fully secure, shared key.

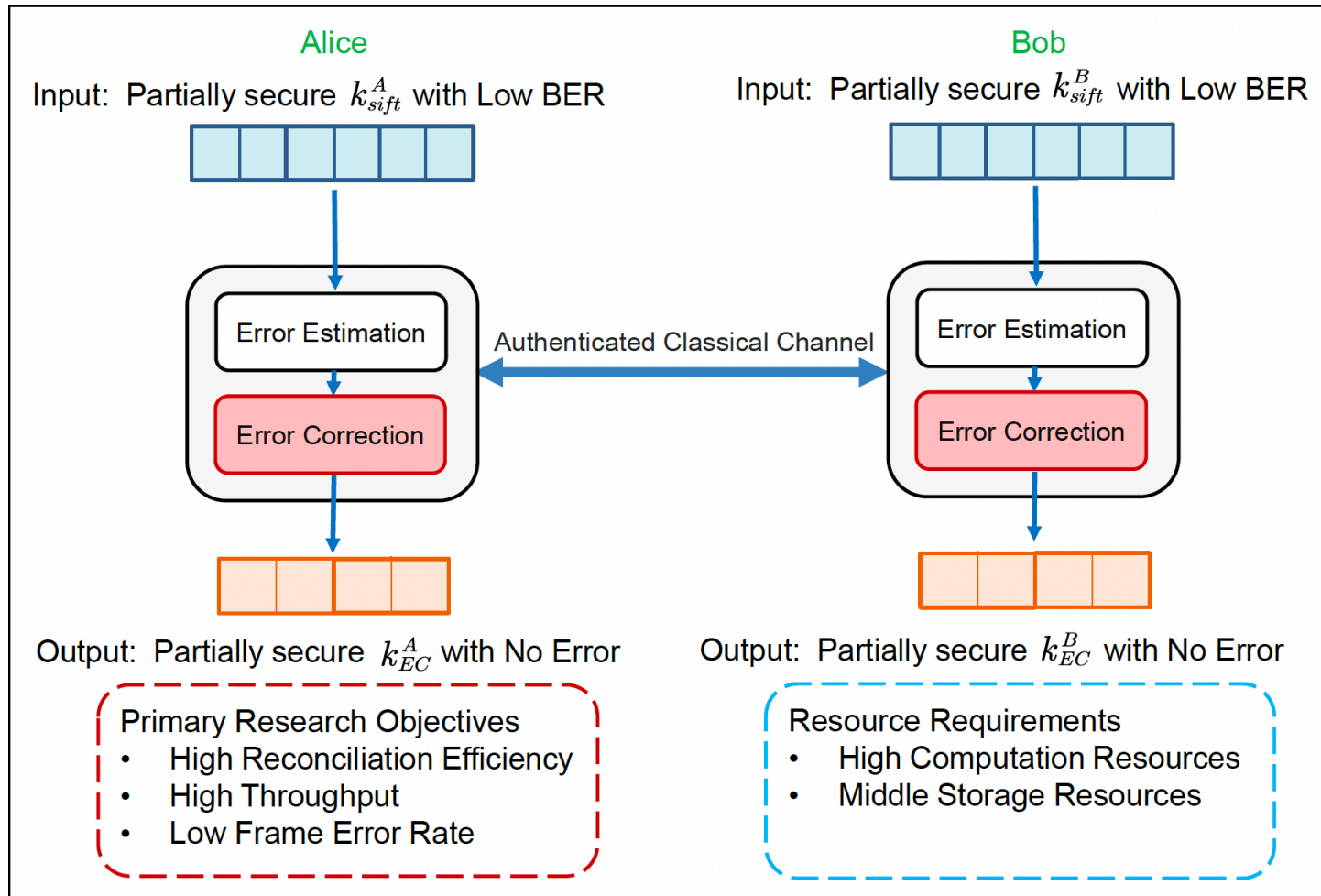


- In all tasks, some bits are sacrificed to ensure correctness and security.
- Efficient algorithms aim to minimize the number of bits discarded.
- Standard techniques are applied in each step.

Sifting



Information Reconciliation



Luo, Yi, et al. "An overview of postprocessing in quantum key distribution." *Mathematics* 12.14 (2024): 2243.



Parameter Estimation

- **Goal:** Estimate the Quantum Bit Error Rate (QBER).
- **Alice and Bob reveal a small subset of bits:**

For Example :

- Alice bits: 101011001
- Bob bits: 101**1**11001

One mismatch bit is observed.

- **Errors arise from:**
 - Fiber attenuation
 - Detector imperfections
- **An eavesdropper introduces disturbances.**
 - According to quantum mechanics:
Information Gain $\Rightarrow$ Disturbance
Large QBER $\Rightarrow$ Possible attack.

$$QBER = \frac{\text{Number of Errors}}{\text{Total Compared Bits}}$$

- **Security Decision**
- if $QBER > QBER_{\text{threshold}}$

Abort Protocol

- ***Else:***

Continue to reconciliation

For the BB84 Protocol, the QBER Limit is $\approx 11\%$



Information Reconciliation

- **Goal:** To transform

Alice: 101101101101

Bob: 101100101101

Bits into

Alice: 101101101101

Bob: 101101101101

without revealing the key.

- Most reconciliation methods use **Parity Information** instead of revealing bits directly.

For Example:

1101

$$1 \oplus 1 \oplus 0 \oplus 1 = 1$$

Only one bit of information leaked

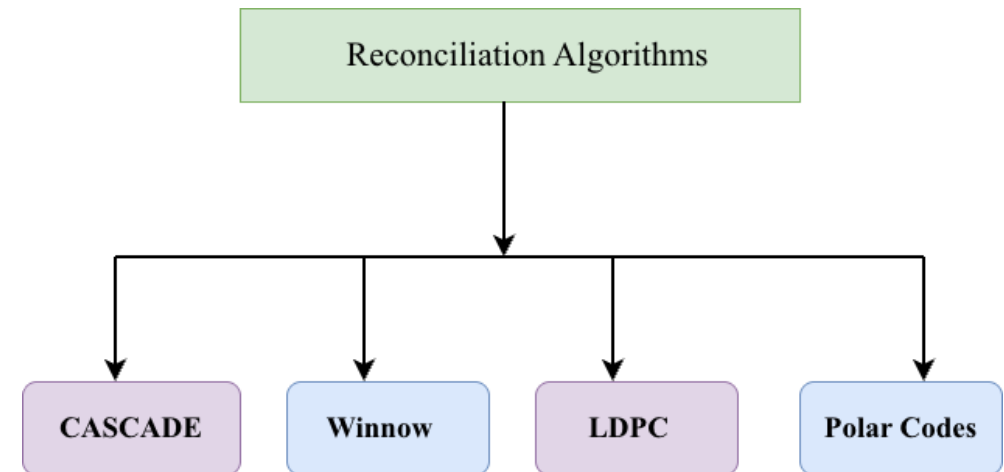
- **Reconciliation challenge**
- Reliability: Correct all errors.
- Security: Leak minimum information.

Trade-off: Reliability $\leftrightarrow$ Information Leakage



Reconciliation Algorithms Evolution

- **CASCADE:** Most influential interactive reconciliation protocol and mostly used in early BB84 implementations.
- **Winnow:** Designed to improve Cascade.
- **LDPC:** Low-Density Parity Check Codes used in fiber QKD, satellite QKD and continuous variable QKD.
- **Polar Codes:** First capacity-achieving coding scheme with low complexity.





Cascade Protocol

- Divide the key into blocks.
- Compare block parity.

Example:

Alice: 01011001

Bob: 01010001

Binary Search Error Localization

0101 | 1001

0101 | 0001

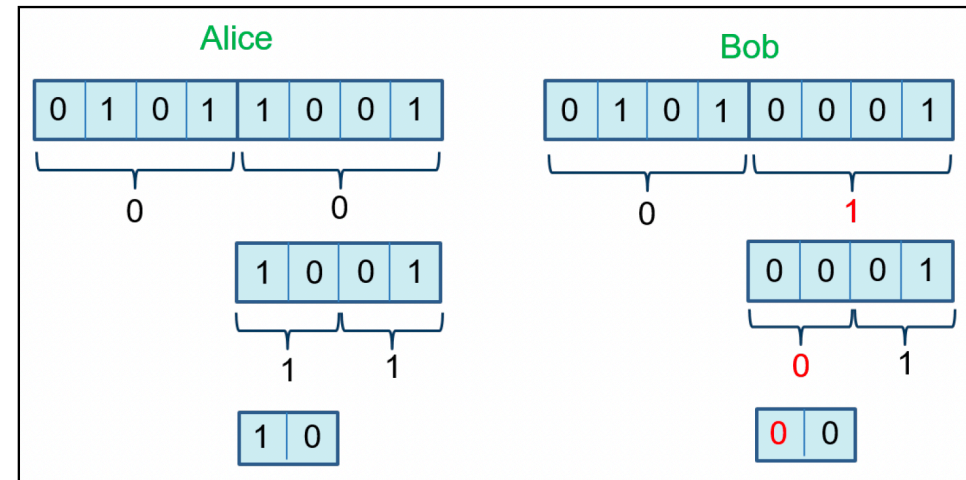
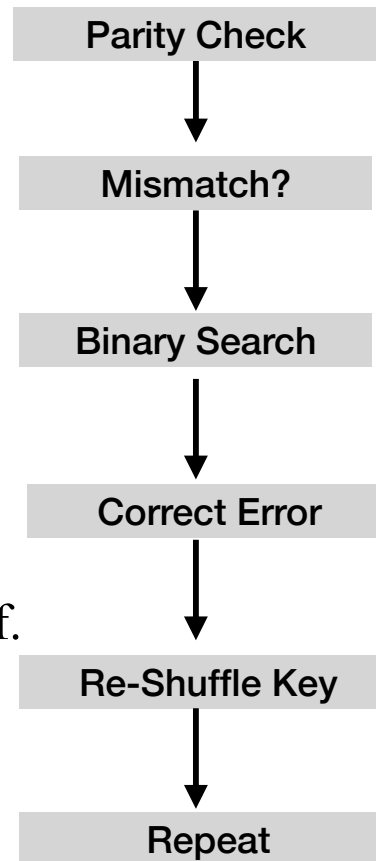
An error was found in the second half.

- **Split again:**

10 | 01

00 | 01

Continue recursively.





Cascade Iterative Passes

- Cascade uses multiples passes:
 - **Pass 1:** Random blocks
 - **Pass 2:** Random permutation
 - **Pass 3:** Different block size
 - **Pass 4:** Repeat

Errors corrected in one pass reveal hidden errors in later passes.

- **Advantages:**

- ✓ Simple
- ✓ Robust
- ✓ Near-optimal efficiency

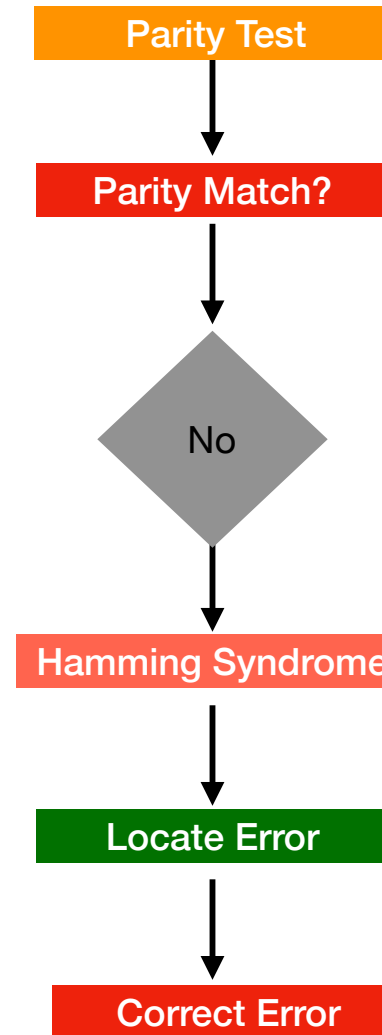
- **Disadvantages:**

- ✗ High interaction
- ✗ Many communication rounds
- ✗ Large latency
- ✗ Not ideal for modern high-rate QKD.

Winnow Algorithm



- It is designed to improve the Cascade protocol.
- **Goal:** Reduce interaction.
- **It Combines:**
 - Parity checks
 - Hamming codes
- **Advantages:**
 - ✓ Faster than Cascade
 - ✓ Lower communication
 - ✓ Reduced leakage
- **Limitations:**
 - ✗ Moderate error tolerance
 - ✗ Less effective at large QBER.





Winnow Algorithm

- $c_A = [1 \ 0 \ 0 \ 1] \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} = 1001100$
- $S_A = 100$ (Sent to Bob)
- $c_B = m_B G = 0001111: S_B = 111$
- $S_A + S_B = 011$: 1st column of $H \rightarrow$ 1st bit is in error

